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Section: BSIT-102I

WW2 - M3U5_M4U1 Assignment #10

M3U5 (File Systems)

1. A physical drive is the actual hardware disk inside your computer. A logical drive is a virtual partition labeled with a drive letter that the operating system uses to organize files. A single physical disk can have multiple logical drives by partitioning it into separate sections to keep personal files separate from system files or to run different operating systems. Conversely, a single logical drive can represent multiple physical devices when using RAID or Windows Storage Spaces to combine multiple physical drives into one larger volume.
2. An IT administrator can find the default location for a user's personal documents at C:\Users\Documents. It is important not to move or delete essential system folders like C:\Windows because they contain core operating system files, drivers, and system utilities needed for Windows to run, and altering them can crash the system or make the computer unbootable.
3. To save a new version of a file without overwriting the original, you would use the Save As command. Deleting files from your main hard drive is safer because they are sent to the Recycle Bin where they can easily be restored, whereas deleting files from a USB flash drive bypasses the Recycle Bin and deletes them permanently.
4. Open File Explorer and navigate to the target folder. In the search bar at the top right, type kind:picture datemodified:last month to filter for images from last month. Next, switch the view layout to Details using the View tab. Finally, click the Size column header to sort the search results from largest to smallest.
5. TIFF is an uncompressed or losslessly compressed image format that preserves full image quality but results in very large file sizes. JPEG uses lossy compression to significantly reduce file size with a minor loss in image quality. Using a ZIP file helps you send multiple files efficiently by bundling them into a single compressed container that takes up less space and sends faster. A compressed file actively shrinks data using compression algorithms and must be extracted before editing, while a regular folder is simply an organizational directory on the storage drive.

M4U1 (Networking Concepts)

1. A LAN (Local Area Network) covers a small local area like a single home or office, whereas a WAN (Wide Area Network) spans large geographical areas and connects multiple smaller LANs. In a home setup, your home network forms a LAN, the internet is a global WAN, and your router connects the two by directing local network traffic to the internet through your ISP.

2. The purpose of a data packet is to break large messages into smaller units for efficient network transmission, allowing data to take available paths and preventing the need to resend entire files if an error occurs. A router forwards data packets between different networks to guide them to their destination. It uses the destination IP address in the packet header along with its internal routing table to determine the best path.
3. An IP address is a logical network address used to locate devices across interconnected networks. A MAC address is a unique physical address assigned to a network interface card for local delivery within a network. The Domain Name System (DNS) translates human-readable domain names like www.google.com into numerical IP addresses so computers know which server to connect to.
4. POP3 downloads emails from the server to a single local device and usually deletes them from the server. IMAP keeps emails on the server and synchronizes message states across all connected devices. For someone using multiple devices, choosing IMAP ensures that read status, folders, and deleted emails stay synchronized across all devices rather than being scattered on one device.
5. An https:// prefix means the connection between the web browser and the server is encrypted using security protocols so transmitted data cannot be read by third parties. SSL/TLS encrypts sensitive data in transit such as passwords and credit card numbers, while a digital certificate verifies the identity of the website to ensure you are communicating with the legitimate owner rather than an impostor.